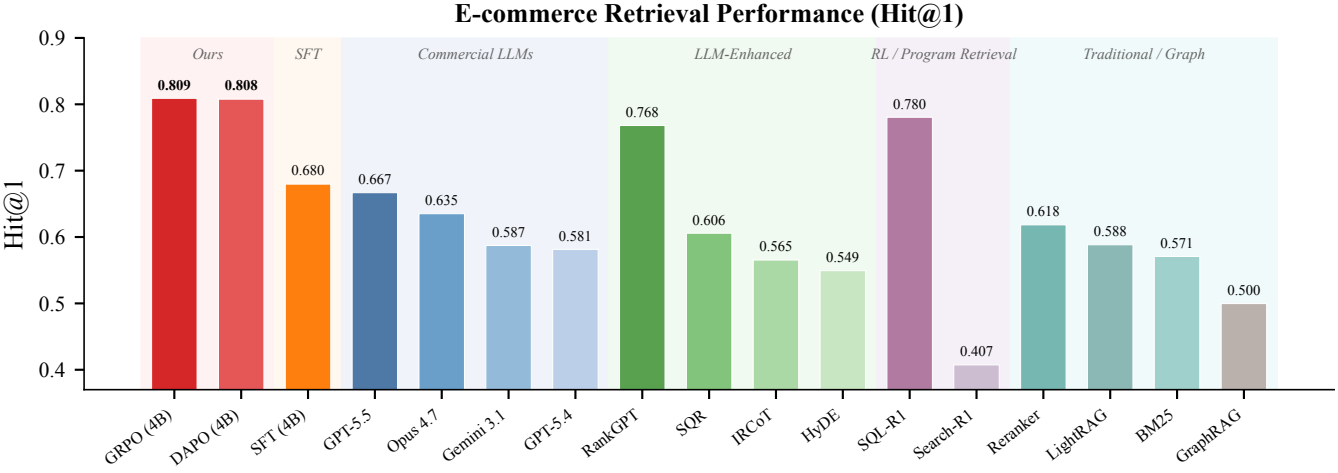

E-commerce Retrieval Performance (Hit@1)
Hit@1
0.9
0.8
0.7
0.6
0.5
0.4
Ours
SFT
Commercial LLMs
LLM-Enhanced
RL / Program Retrieval
Traditional / Graph
0.809
0.808
0.680
0.667
0.635
0.587
0.581
0.768
0.606
0.565
0.549
0.780
0.407
0.618
0.588
0.571
0.500
GRPO (4B)
DAPO (4B)
SFT (4B)
GPT-5.5
Opus 4.7
Gemini 3.1
GPT-5.4
RankGPT
SQR
IRCoT
HyDE
SQL-R1
Search-R1
Reranker
LightRAG
BM25
GraphRAG